

Mini Project

Mathematics for New Technologies in Finance
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Part I

1. Θ_0 is the set of admissible trading strategies determined by the gradient of a $C^2(E)$ potential function φ . The strategies in this set satisfy:
 - Markovian property: Dependency on the current price X_t , but not on the path of X before time t .
 - Predictability: Since $\varphi \in C^2(E)$, $\nabla\varphi(X_t)$ is continuous and $\mathcal{F}_t$ -measurable, making the strategy predictable.
 - The weights are determined by the gradient: investors hold more of an asset if the potential increases in its direction.
 - φ satisfies a specific integrability condition necessary for the ergodic property.
2. Let $\theta_0 \in \Theta_0$ and $\varphi \in C^2(E)$ be its potential function. As X is a continuous semimartingale and $\varphi \in C^2(E)$, we apply Itô's formula

$$d\varphi(X_t) = \nabla\varphi(X_t)^\top dX_t + \frac{1}{2} \sum_{i,j=1}^d \frac{\partial^2 \varphi(X_t)}{\partial X_t^i \partial X_t^j} d\langle X^i, X^j \rangle_t = \nabla\varphi(X_t)^\top dX_t + \frac{1}{2} \text{Tr}(c \nabla^2 \varphi(X_t)) dt, \quad (1)$$

and denote $M_T = \int_0^T \theta_t^\top dX_t$, so $\langle M \rangle_t = \nabla\varphi(X_t)^\top \langle X \rangle_t \nabla\varphi(X_t) dt = \nabla\varphi(X_t)^\top c \nabla\varphi(X_t) dt$. Since X is a continuous semimartingale and $\nabla\varphi(X)$ is predictable and continuous, M is a continuous semimartingale. It is direct to see M is null at 0. This is why $V_T^{\theta_0} := \varepsilon(M)_T = e^{M_T - \frac{1}{2} \langle M \rangle_T}$, and thus

$$\boxed{\log(V_T^{\theta_0})} = M_T - \frac{1}{2} \langle M \rangle_T = \varphi(X_T) - \varphi(X_0) - \frac{1}{2} \int_0^T \text{Tr}(c \nabla^2 \varphi(X_t)) dt - \frac{1}{2} \int_0^T \nabla\varphi(X_t)^\top c \nabla\varphi(X_t) dt. \quad (2)$$

From (2) and the ergodic behavior of X one gets

$$\liminf_{T \rightarrow \infty} \frac{1}{T} \log(V_T^{\theta_0}) = \boxed{-\frac{1}{2} \int_E [\text{Tr}(c \nabla^2 \varphi(x)) + \nabla\varphi(x)^\top c \nabla\varphi(x)] p(x) dx = g(\theta_0; \mathbb{P})}, \quad (3)$$

so $g(\theta_0; \mathbb{P})$ is independent of $\mathbb{P} \in \Pi \forall \theta_0 \in \Theta_0$, as $\liminf_{T \rightarrow \infty} \frac{1}{T} \log(V_T^{\theta_0})$ depends on c , $\varphi(x)$ and $p(x)$, which are not dependent on $\mathbb{P}$. That is, Θ_0 has the growth rate invariance property over Π . In the terms of our problem, this means we reduce it to a pure maximization problem, as the problem now reads as

$$\sup_{\theta_0 \in \Theta_0} \inf_{\mathbb{P} \in \Pi} g(\theta_0; \mathbb{P}) = \sup_{\theta_0 \in \Theta_0} g(\theta_0; \mathbb{P}). \quad (4)$$

3. As $\text{Tr}(A(x) \nabla^2 e^{\varphi(x)}) = \sum_{i,j=1}^d A_{ij}(x) \partial_{ij} e^{\varphi(x)}$, we have

$$\begin{aligned} \boxed{J} &= \int_E \frac{\text{Tr}(A(x) \nabla^2 e^{\varphi(x)})}{e^{\varphi(x)}} dx = - \sum_{i,j=1}^d \int_E (\partial_i A_{ij}(x) \partial_j \varphi(x) - \partial_i \varphi(x) A_{ij}(x) \partial_j \varphi(x)) dx \\ &= \int_E \nabla\varphi(x)^\top A(x) (\nabla\varphi(x) - A^{-1}(x) \text{div} A(x)) dx, \end{aligned} \quad (5)$$

where in the second equality, we used integration by parts and that φ is compactly bounded.

4. In our case, since $A : E \rightarrow \mathbb{R}^{d \times d}$ (goes from a d -dimensional space to a $d \times d$ -dimensional space), $\text{div} A(x)$ simply is a vector with entries

$$(\text{div} A(x))_i = \sum_{j=1}^d \partial_j A_{ij}(x) = \sum_{j=1}^d \partial_j (c_{ij} p(x)) = \sum_{j=1}^d c_{ij} \partial_j p(x) \quad \longrightarrow \quad \boxed{\text{div} A(x)} = \nabla A(x) = \boxed{c \nabla p(x)}. \quad (6)$$

Furthermore, it is straightforward that $A^{-1}(x) = \frac{c^{-1}}{p(x)}$, and therefore

$$\boxed{A^{-1}(x) \nabla A(x)} = \frac{c^{-1}}{p(x)} c \nabla p(x) = \frac{\nabla p(x)}{p(x)} \boxed{= \nabla \log p(x)}. \quad (7)$$

5. From exercise 3 and $A(x) = cp(x)$ we obtain $g(\theta_0; \mathbb{P}) = -\frac{1}{2}J$. From this, the result from exercise 4 and $\nabla \log p(x) = -\Sigma^{-1}x$, for $\theta_0 \in \Theta_0$ one can rewrite the optimization problem as

$$\sup_{\theta_0 \in \Theta_0} g(\theta_0; \mathbb{P}) = \sup_{\theta_0 \in \Theta_0} -\frac{1}{2} \int_E p(x) [\nabla \varphi(x)^\top c \nabla \varphi(x) - x^\top \Sigma^{-1} c \nabla \varphi(x)] dx. \quad (8)$$

We can clearly see that as $g(\theta_0; \mathbb{P})$ **quadratic** in $\nabla \varphi$ in the first term, and **linear** in $\nabla \varphi$ in the second term, i.e. our maximization problem is LQ. This suggests the optimizer is a quadratic function of x . Thus, we make the ansatz $\varphi^*(x) = \frac{1}{2}x^\top Qx$ (the $\frac{1}{2}$ is added for the simplicity of the gradient $\nabla \varphi^*(x) = Qx$) for some symmetric $Q \in \mathbb{S}^d$. Then,

$$g(\theta_0; \mathbb{P}) = -\frac{1}{2} \int_E p(x) x^\top [QcQ + \Sigma^{-1}cQ] x dx = -\frac{1}{2} \left(\text{Tr}(QcQ\Sigma) + \text{Tr}(cQ) \right), \quad (9)$$

where we used the Gaussian second momentum formula in the last equality. Taking the derivative with respect to Q and setting it to zero one obtains $2cQ\Sigma + c = 0$, so $Q = -\frac{1}{2}\Sigma^{-1}$ and the optimal φ is

$$\boxed{\varphi^*(x) = -\frac{1}{4}x^\top \Sigma^{-1}x}. \quad (10)$$

6. Considering result (10) and $\theta_t^* = \nabla \varphi^*(X_t)$, the robust-optimal trading strategy is

$$\boxed{\theta_t^* = -\frac{1}{2}\Sigma^{-1}X_t} \quad (11)$$

$\forall t \geq 0$. This strategy maximizes the asymptotic growth rate over all admissible gradient strategies, and since $g(\theta_0; \mathbb{P})$ is independent of $\mathbb{P} \in \Pi$, it is automatically robust against model uncertainty.

7. Considering the results from exercise 5, the optimal growth rate is

$$\boxed{g^*(\theta^*; \mathbb{P})} = \frac{1}{8} \int_E p(x) x^\top [\Sigma^{-1}c\Sigma^{-1}] x dx = \frac{1}{8} \mathbb{E}[X^\top \Sigma^{-1}c\Sigma^{-1}X] = \boxed{\frac{1}{8} \text{Tr}(\Sigma^{-1}c)}. \quad (12)$$

From this result we observe that the robust-optimal growth rate depends only on instantaneous covariance matrix c and the invariant covariance matrix Σ , not under any variable depending on $\mathbb{P} \in \Pi$, so the optimal strategy is genuinely robust under model uncertainty.

Part II

1. The general d -dimensional OU dynamics of X is

$$dX_t = (\mu(t) - B(t)X_t)dt + \sigma(t)dW_t, \quad (13)$$

for $\mu(t) \in \mathbb{R}^d$ a constant drift factor, $B(t) \in \mathbb{R}^{d \times d}$ the mean-reversion matrix and $\sigma(t) \in \mathbb{R}^{d \times d}$ the volatility matrix. Nature will perturb the drift part, which in our case is $B(t) = B$ ($\mu(t) = 0$) while respecting the stationarity distribution constraint, as $\sigma(t) = \sigma$ is fixed by the instantaneous covariance matrix $c = \sigma\sigma^\top$.

2. In our case, the Fokker-Planck equation reads

$$0 = -\nabla [Bxp(x)] + \nabla^2 \left[\frac{1}{2}cp(x) \right] = \nabla \left[-Bxp(x) + \frac{1}{2}c\nabla p(x) \right] = \nabla \left[\left(-B - \frac{1}{2}c\Sigma^{-1} \right) xp(x) \right], \quad (14)$$

where in the last equality we used $\nabla p(x) = -\Sigma^{-1}xp(x)$. Using the symmetry of c and Σ , equation (14) can only hold if

$$\left(-B - \frac{1}{2}c\Sigma^{-1} \right) = 0 \quad \longrightarrow \quad B\Sigma = -\frac{1}{2}c, \quad \Sigma B^\top = (B\Sigma)^\top = -\frac{1}{2}c^\top = -\frac{1}{2}c. \quad (15)$$

In other words, to maintain a valid stationary distribution Nature must satisfy the Lyapunov-type constraint

$$\boxed{B\Sigma + \Sigma B^\top = -c}. \quad (16)$$

3. The matrix $B\Sigma + \Sigma B^\top \in \mathbb{R}^{d \times d}$ is symmetric, therefore, it has $\frac{d(d+1)}{2}$ independent entries. This means that (16) imposes $\frac{d(d+1)}{2}$ independent constraints on B . As B has a total of d^2 independent entries, the number of entries without a constraint (that is, the degrees of freedom) which Nature can modify is

$$d^2 - \frac{d(d+1)}{2} = \boxed{\frac{d(d-1)}{2}}. \quad (17)$$

4. **Agent:** They have trading strategy $\theta_t = \nabla \varphi(X_t)$, so we need to parametrize φ . We parametrize φ as a Multilayer Perceptron with 3 hidden layers, each with $m = 64$ neurons

$$\varphi_\alpha : \mathbb{R}^d \xrightarrow{\text{linear}} \mathbb{R}^m \xrightarrow{\tanh} \mathbb{R}^m \xrightarrow{\tanh} \mathbb{R}^m \xrightarrow{\text{linear}} \mathbb{R} \quad (18)$$

with tanh activation functions, which guarantee $\varphi_\alpha \in C^2$. The agent wants to maximize $\hat{g}(\alpha; \beta)$, so the loss function is $-\hat{g}(\alpha; \beta)$.

Adversary: Nature modifies $B \in \mathbb{R}^{d \times d}$ under (16). We parametrize this constraint as $B_\beta = -\frac{1}{2}c\Sigma^{-1} + S_\beta\Sigma^{-1}$, where $S_\beta \in \mathbb{R}^{d \times d}$ is skew-symmetric. We parametrize the network, which learns $\frac{d(d-1)}{2}$ entries of S_β , as a learnable parameter vector with no network layers a part from B_β reconstruction

$$\beta \in \mathbb{R}^{\frac{d(d-1)}{2}} \mapsto S_\beta \in \mathbb{R}^{d \times d} \mapsto B_\beta \in \mathbb{R}^{d \times d}. \quad (19)$$

The adversary wants to minimize $\hat{g}(\alpha; \beta)$, so the loss function is $\hat{g}(\alpha; \beta)$.

Minimax game: $\max_\alpha \min_\beta \hat{g}(\alpha; \beta)$, where $\hat{g}$ is the Monte Carlo estimator

$$\hat{g}(\alpha; \beta) = \frac{1}{N} \sum_{i=1}^N \left(-\nabla \varphi_\alpha(x^{(i)})^\top B_\beta x^{(i)} - \frac{1}{2} \nabla \varphi_\alpha(x^{(i)})^\top c \nabla \varphi_\alpha(x^{(i)}) \right). \quad (20)$$

5. The code python file can be found in MNF Mini Project Github Repository. For handling instabilities: we used gradient clipping to both the agent and the adversary, to prevent exploding gradients from the backpropagation through the quadratic term; we used a large batch size that reduces Monte Carlo variance and we used float64 precision to avoid numerical noise.

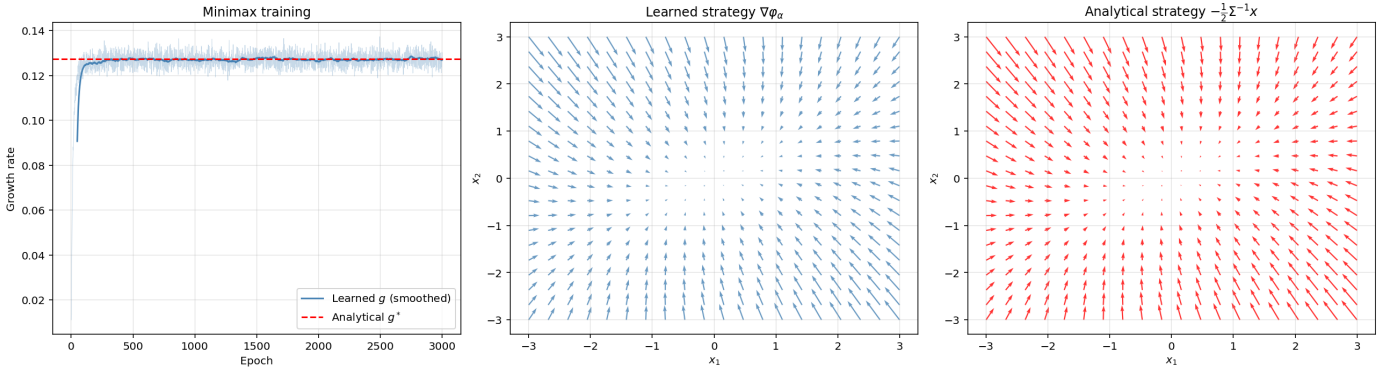


Figure 1: Robust-optimal results: (Left) Convergence of the growth rate $\hat{g}$ to the analytical optimum g^* ; (Center) Learned strategy θ ; (Right) Analytical strategy θ^* .

6. From the training curve in the first image, we can see that the learned growth rate converges in approximately $\lesssim 200$ epochs and stabilizes around the analytical optimum $g^* \approx 0.127273$. Oscillations are expected and caused by the Monte Carlo variance. From the strategy plots in the second and third images, one sees the resemblance of the learned strategy to the analytical one. The arrows point towards the origin, reflecting the off-diagonal structure of Σ , and has the expected behavior.
7. Both approaches lead to consistent results:

Growth rate: The learning problem's growth rate estimator converges to the optimal growth rate found in Part I.

Optimal strategy: Similarly, the learned strategy almost recovers the analytical one.

Nature's role: The most important finding is that in the learning problem we found that the Nature's gradient with respect to β is effectively zero throughout training. This means that Nature cannot influence $\hat{g}$ by perturbing B_β . This is empirical confirmation of the growth rate invariance property derived in Part I.